

Lecture # 10

MM 318: Multicomponent Alloys

Dr. Sumanta Samal

Contact Details

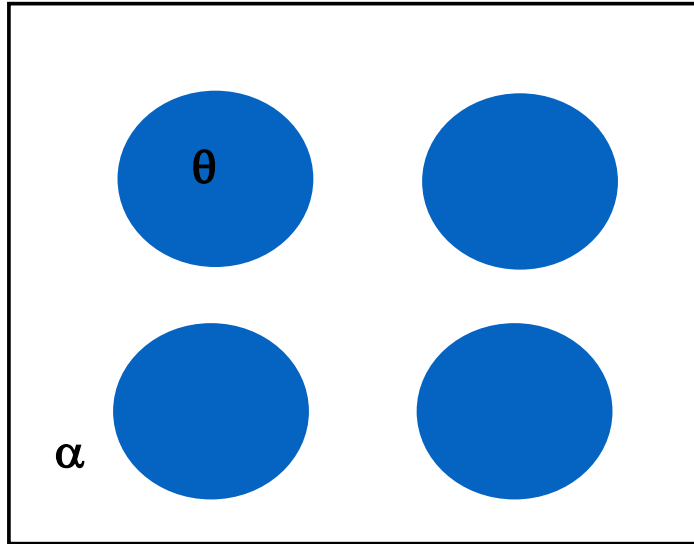
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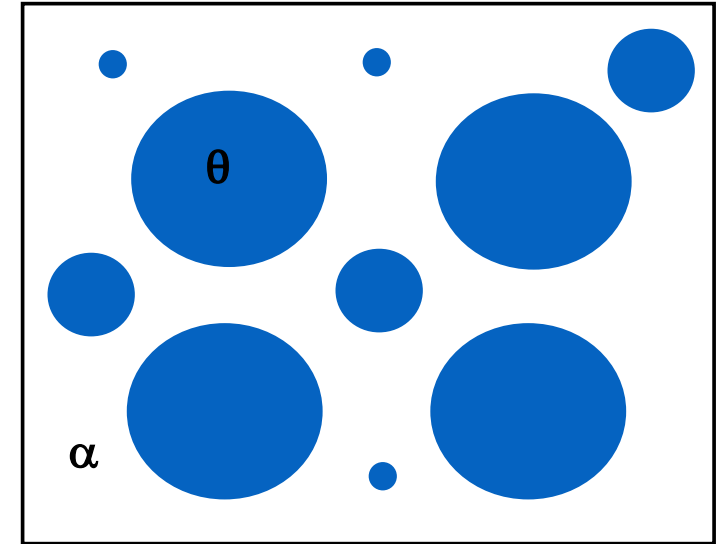
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Particle Coarsening (or Grain Growth or Ostwald Ripening)

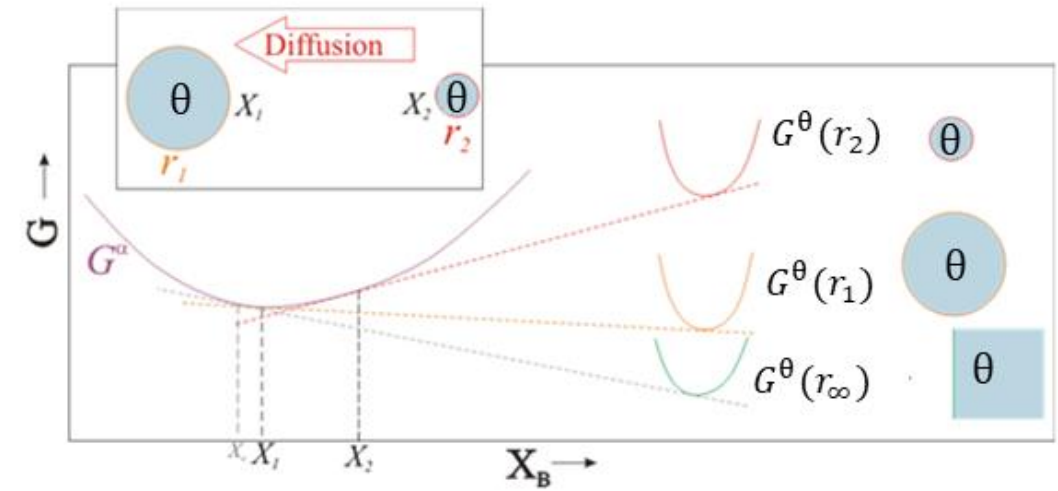
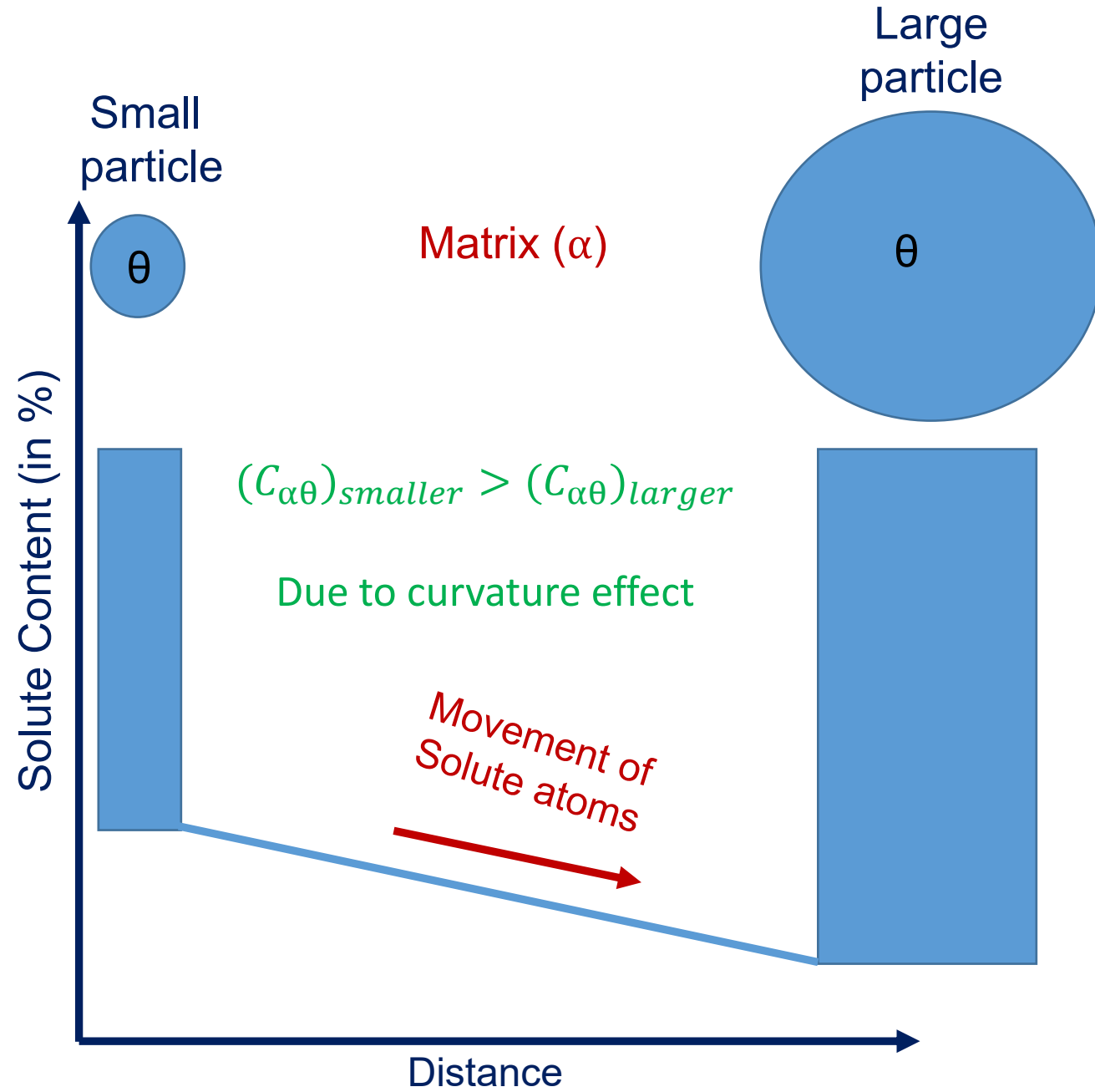


No growth
with particles of same size

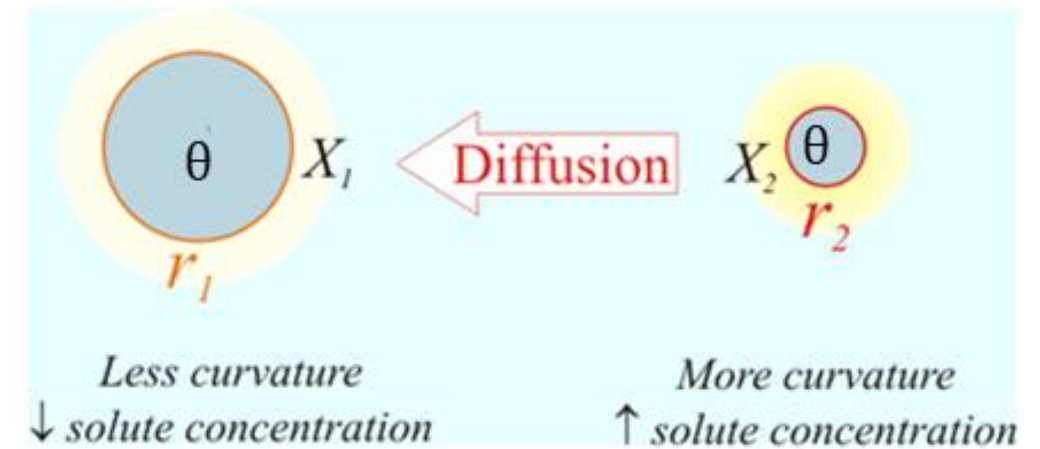


Growth occurs
with particles of different size

Particle coarsening (or grain growth)

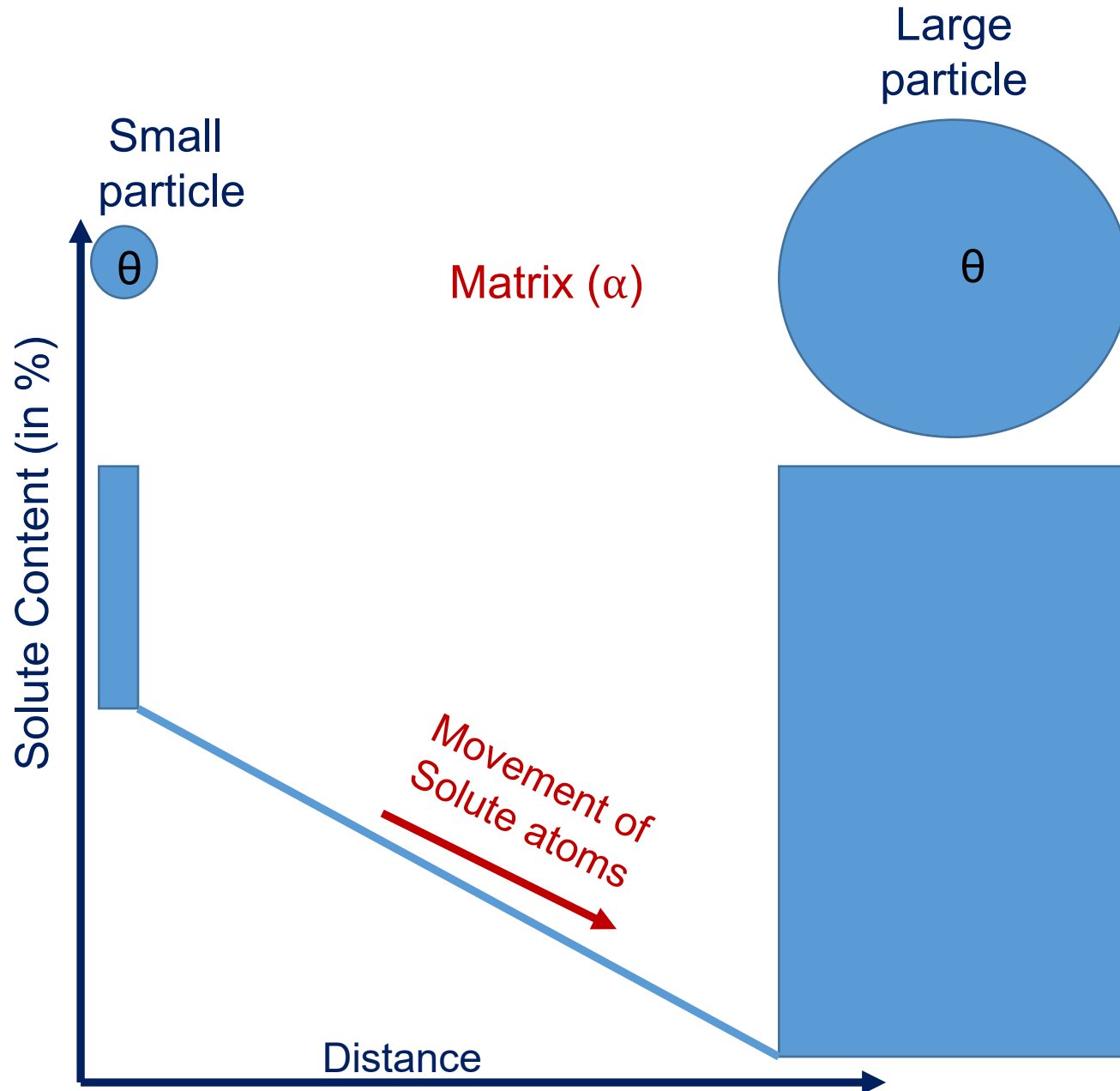


$$X_1 = (C_{\alpha\theta})_{larger} \text{ and } X_2 = (C_{\alpha\theta})_{smaller}$$



Gibbs-Thomson effect

Particle coarsening (or grain growth)



- There will be a range of particle sizes due to time of nucleation and rate of growth.
- As the curvature increases the solute concentration (X_B) in the matrix adjacent to the particle increases
- Concentration gradients are setup in the matrix → solute diffuses from near the small particles towards the large particles
 - ⇒ small particles shrink and large particles grow
- ⇒ with increasing time * Total number of particles decrease
 - * Mean radius (r_{avg}) increases with time

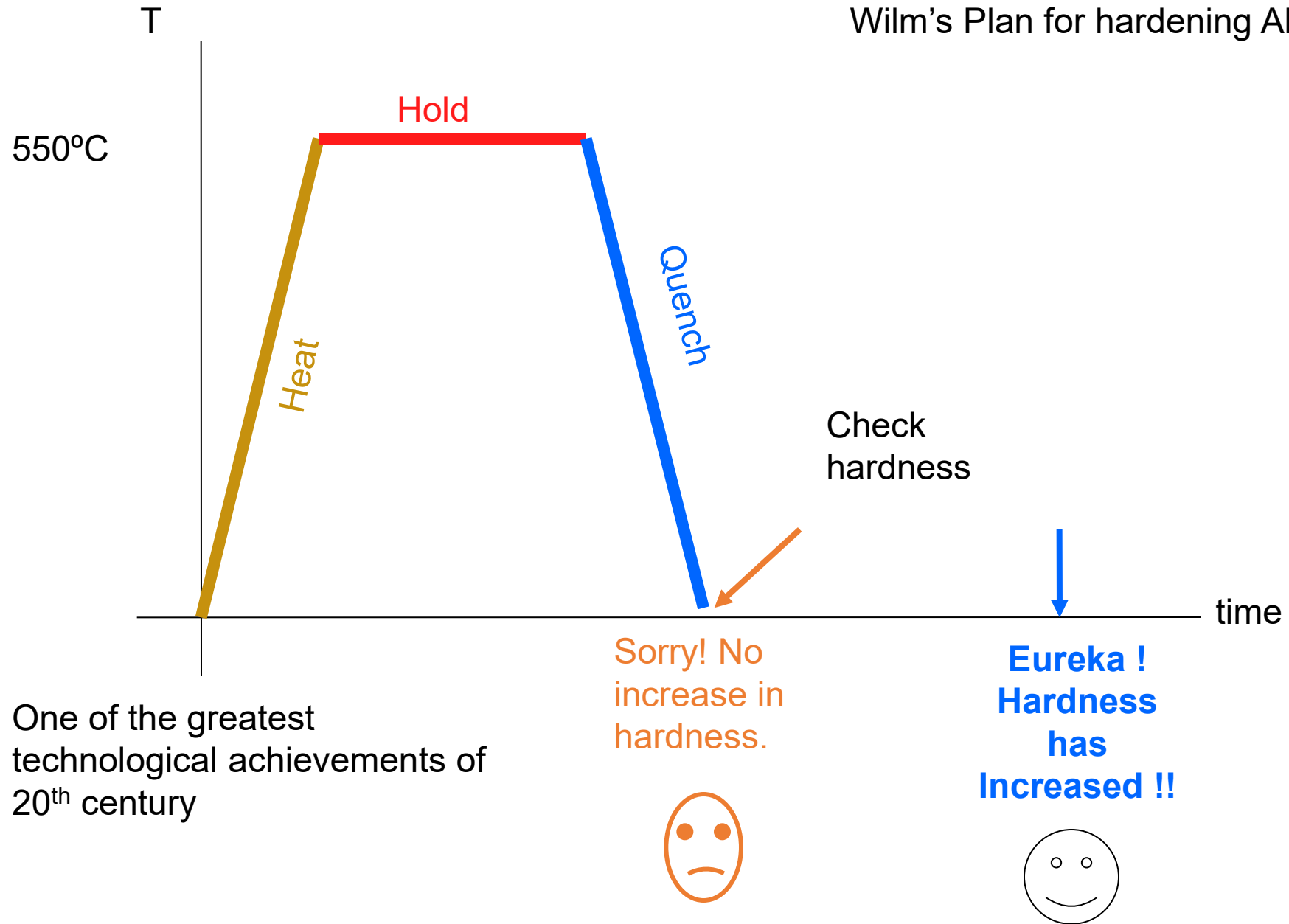
Driving force

$$\Delta C = (C_{\alpha\theta})_{smaller} - (C_{\alpha\theta})_{larger}$$

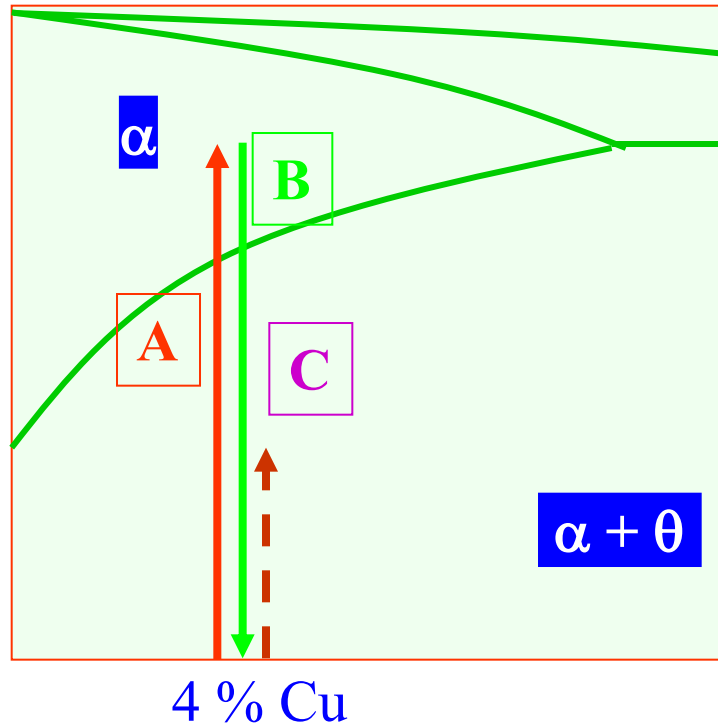
- Driving force increases w.r.t. time

Diffusion of solute during precipitation

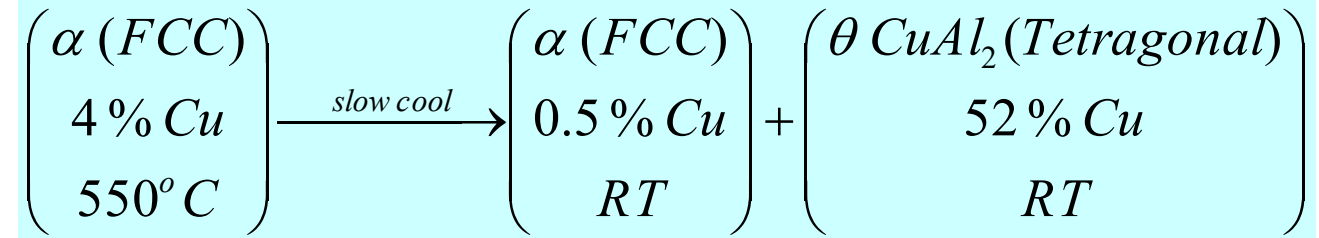
Wilm's Plan for hardening Al-4%Cu alloy



To obtain a fine distribution of precipitates the cycle $A \rightarrow B \rightarrow C$ is used



Note: Treatments A, B, C are for the same composition



- $\alpha \rightarrow \alpha + \theta$
- Slow equilibrium cooling gives rise to coarse θ precipitates which is not good in impeding dislocation motion.*

A Heat (to 550°C) $\rightarrow$ solid solution α

B

Quench (to RT) $\rightarrow$

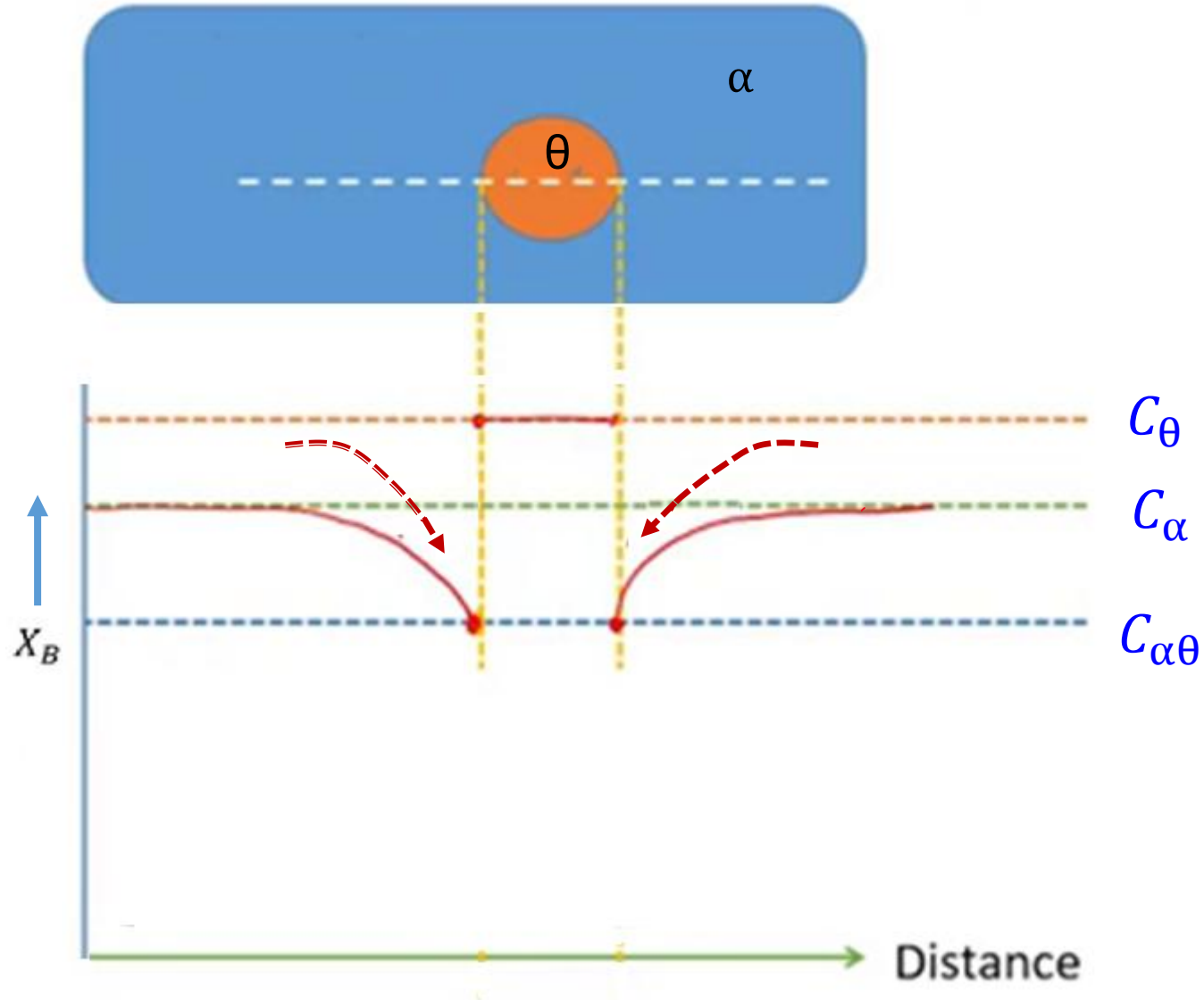
supersaturated solution

Increased vacancy concentration

C

Age (reheat to 200°C) $\rightarrow$ fine precipitates

Composition profiles during precipitation



$C_{\alpha\theta}$: Composition of α equilibrium with θ

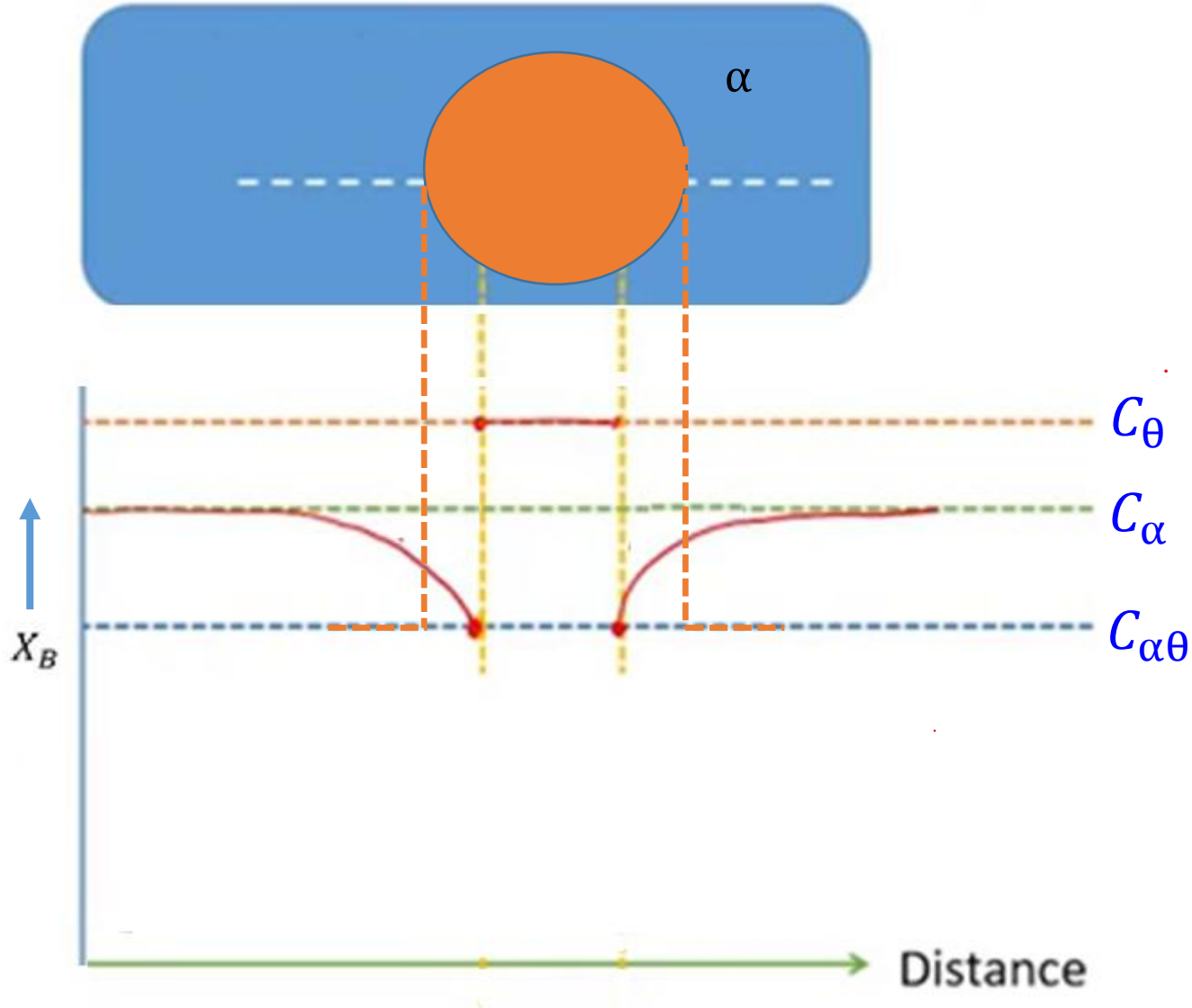
Fick's 1st Law: $J = -D \frac{\partial C}{\partial x}$

Driving force

$$\Delta C = (C_\alpha - C_{\alpha\theta})$$

- At particular temperature, $C_{\alpha\theta}$ is fixed .

Composition profiles during precipitation

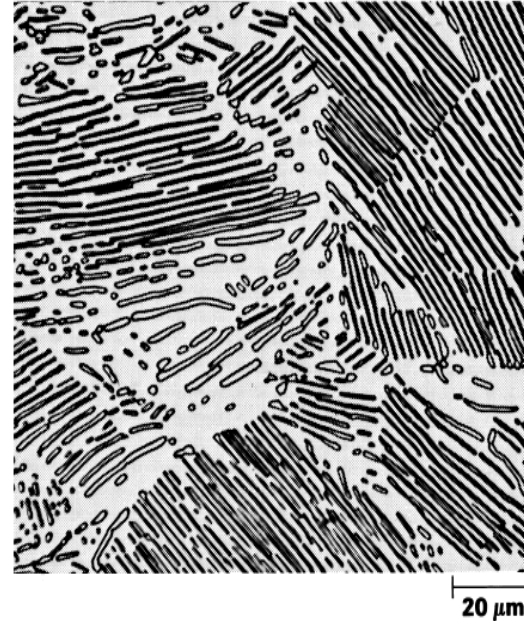


$$\Delta C = (C_\alpha - C_{\alpha\theta})$$

- With increase in time, Diffusion takes place and hence C_α will decrease
- Driving force decreases w.r.t. time

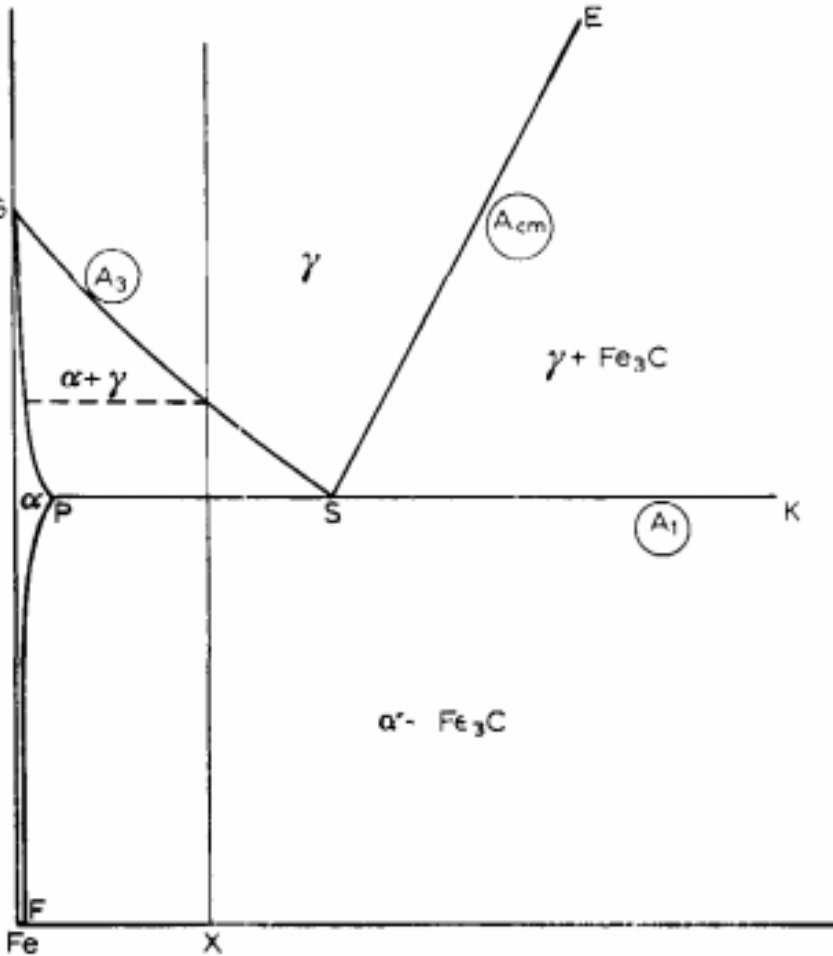
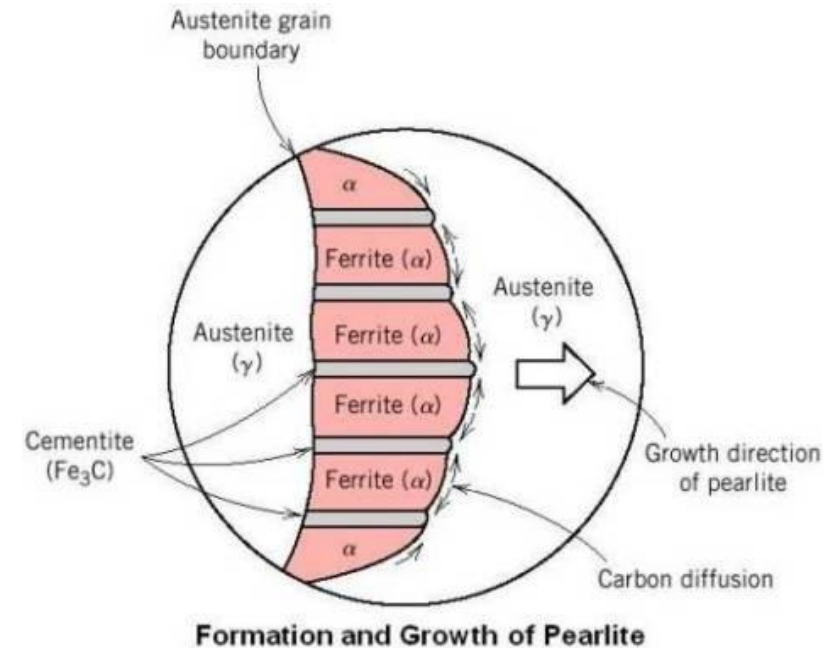
Diffusion takes place until $\Delta C = 0$

Eutectoid reaction



Eutectoid steel

$\alpha + Fe_3C$ (Pearlite)



The eutectoid region of the Fe-Fe₃C system

Driving force: $\Delta C = C_{\gamma\alpha} - C_{\gamma Fe_3C}$

- **Three interfacial energy:**

$\gamma_{\gamma\alpha}, \gamma_{\gamma Fe_3C}, \gamma_{\alpha Fe_3C}$

- **For lamellar growth:**

Interfacial energy between α -phase and Fe_3C phase ($\gamma_{\alpha Fe_3C}$) is lower as compared to $\gamma_{\gamma\alpha}, \gamma_{\gamma Fe_3C}$,

Metallic Glasses

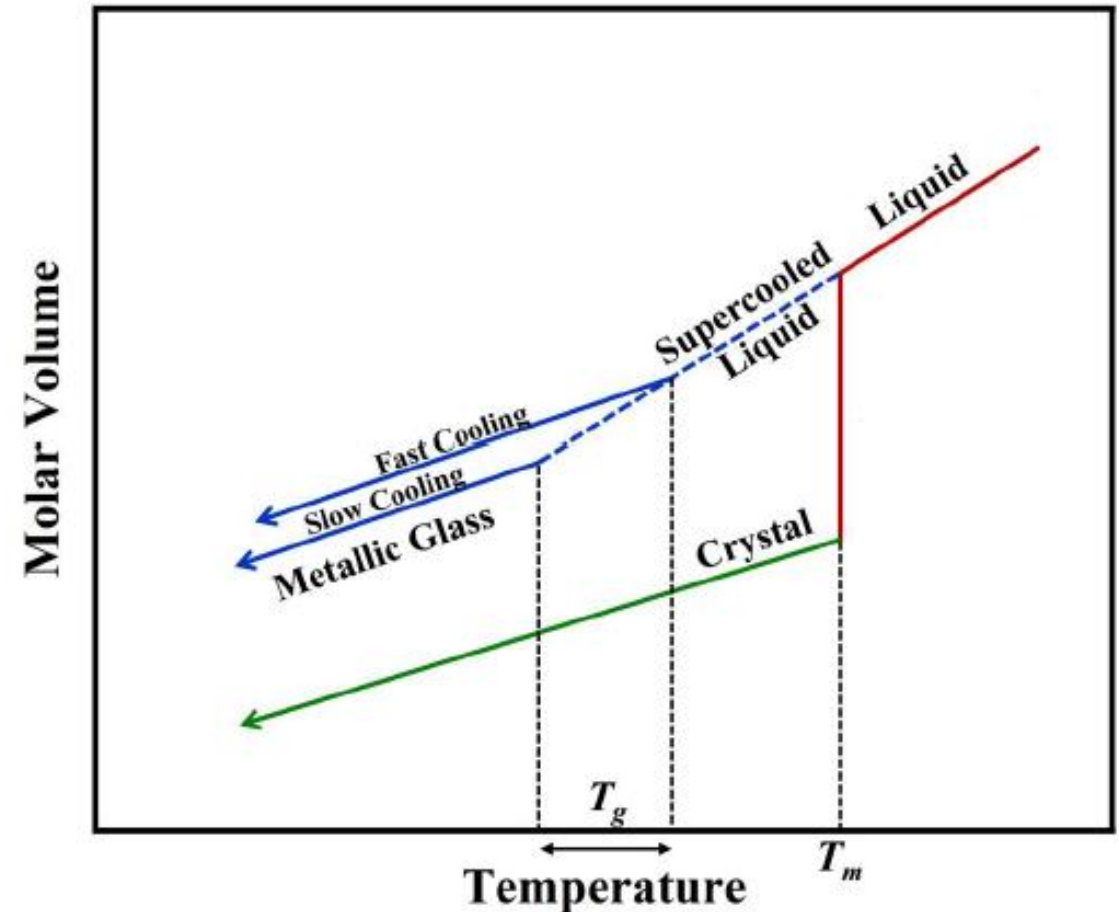
$$S = \int \frac{C_p}{T} dT \Rightarrow \frac{dS}{dT} = \frac{C_p}{T}$$

$$(C_p)_L > (C_p)_S \Rightarrow \left(\frac{dS}{dT}\right)_L > \left(\frac{dS}{dT}\right)_S$$

- Entropy of liquid (S_L) is greater than Entropy of solid (S_S)
- At glass transition temperature (T_g) of ideal glass (= Kauzmann temperature T_k)

$$\Delta S = S_L - S_S = 0$$

- Below T_g ; $S_S > S_L$; which is absurd (or paradox); known as **Kauzmann paradox**
- Liquid freezes and form glass (**configurational frozen supercooled liquid**)
- For alloy, Glass transition temperature (T_g) is higher than pure metal

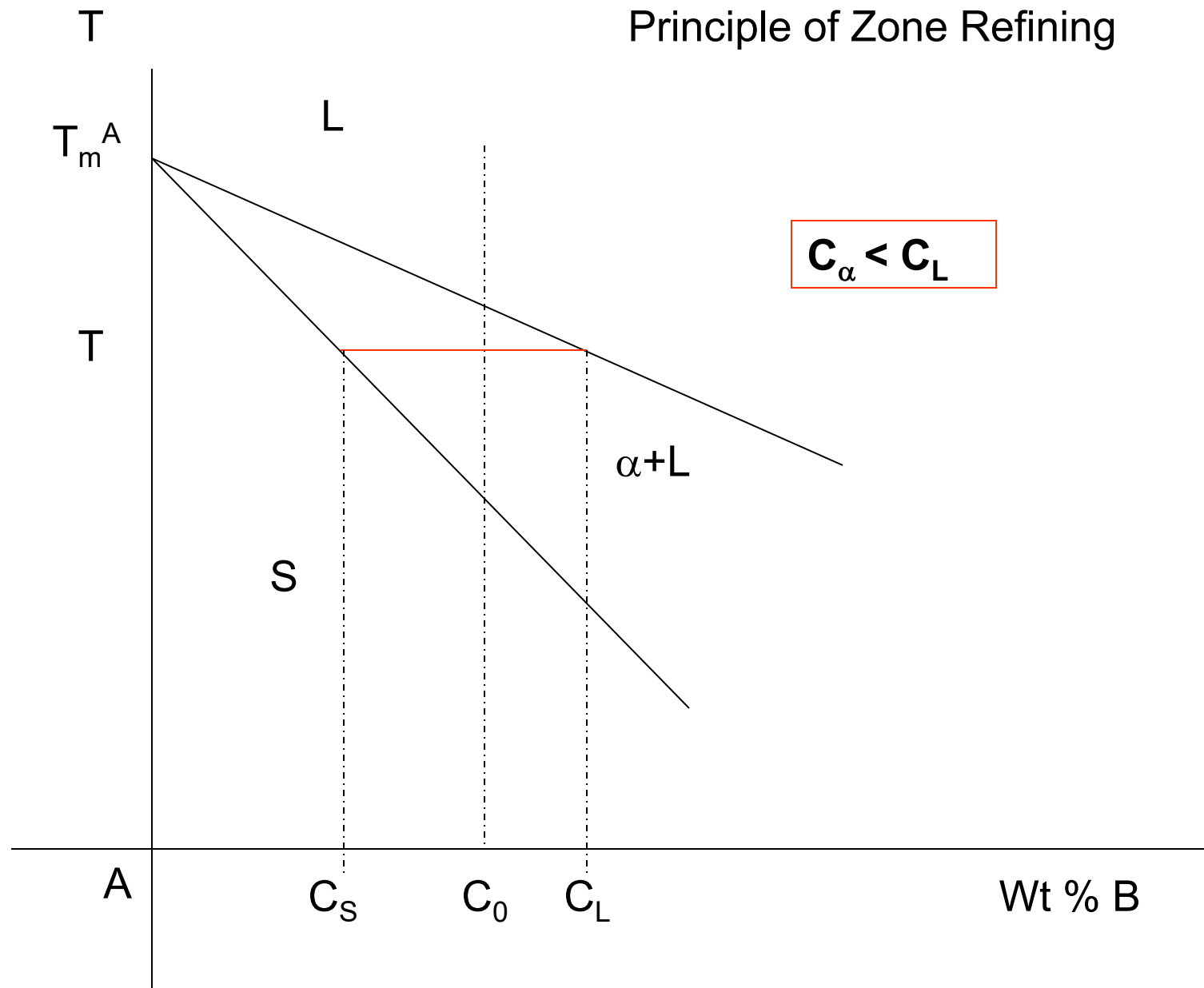


- Molar volume of glass > Molar volume of crystal $\Rightarrow$ Free volume in glass
- $T_g \Rightarrow$ the temperature where atomic configuration is frozen into glass
- Viscosity keeps increasing with a decrease in temperature

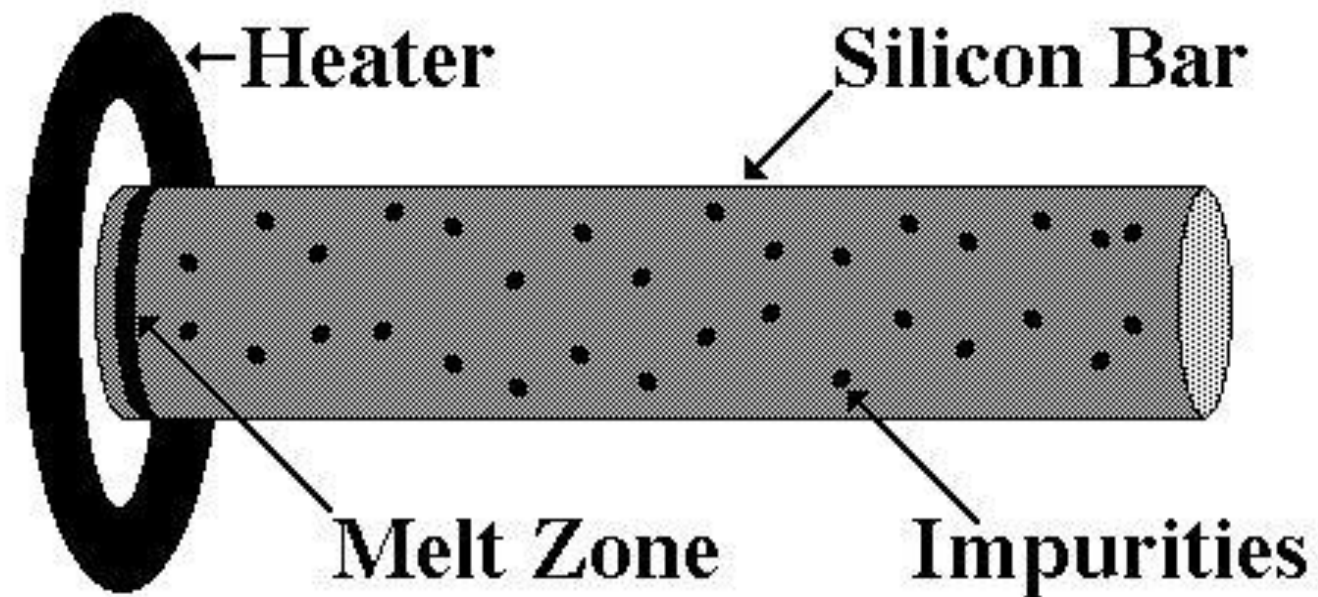
Zone Refining

How is ultrapure Si for computer chips produced?

Principle of Zone Refining



Zone-Refining



Zone-Refining



The heater moves along the bar sweeping impurities along with the melt zone. Repeated passes of the heater end up concentrating all of the impurities in one end, which is then discarded.

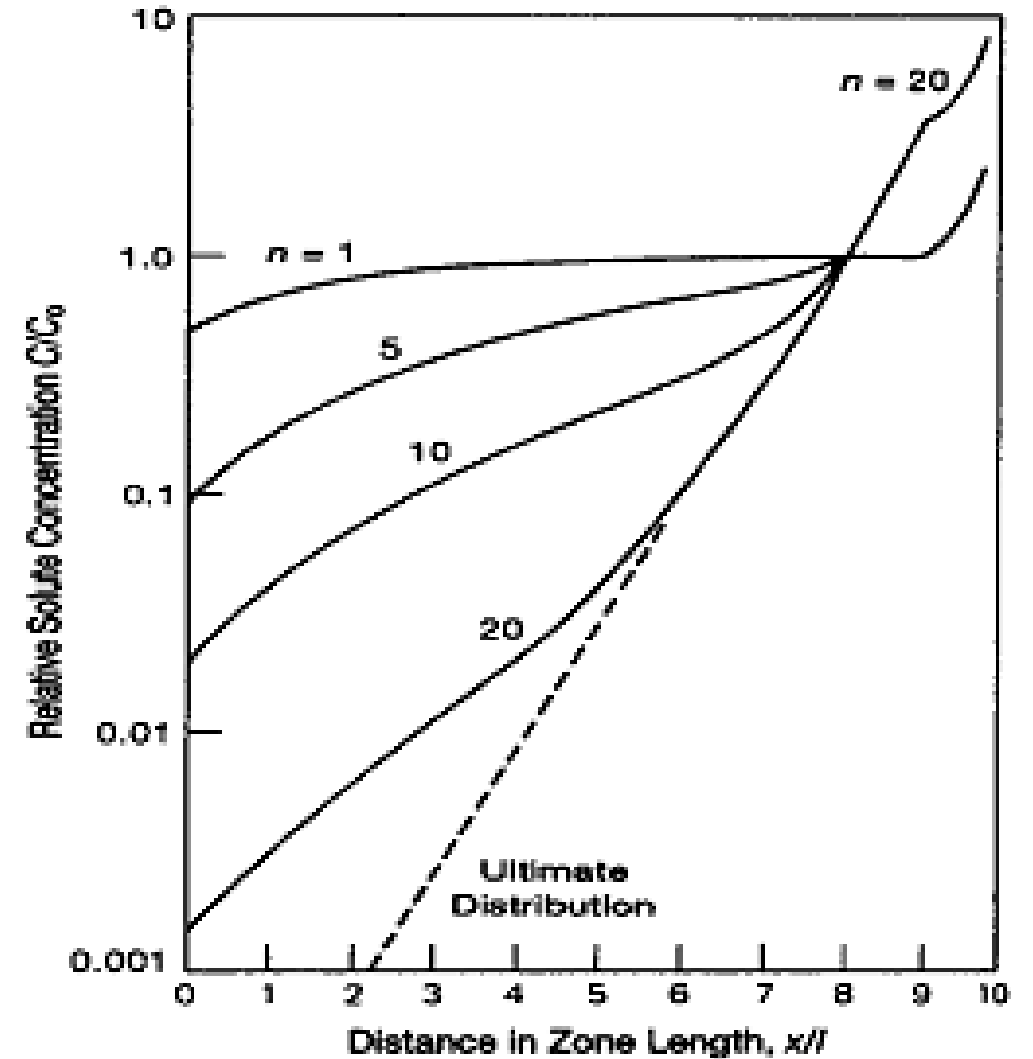
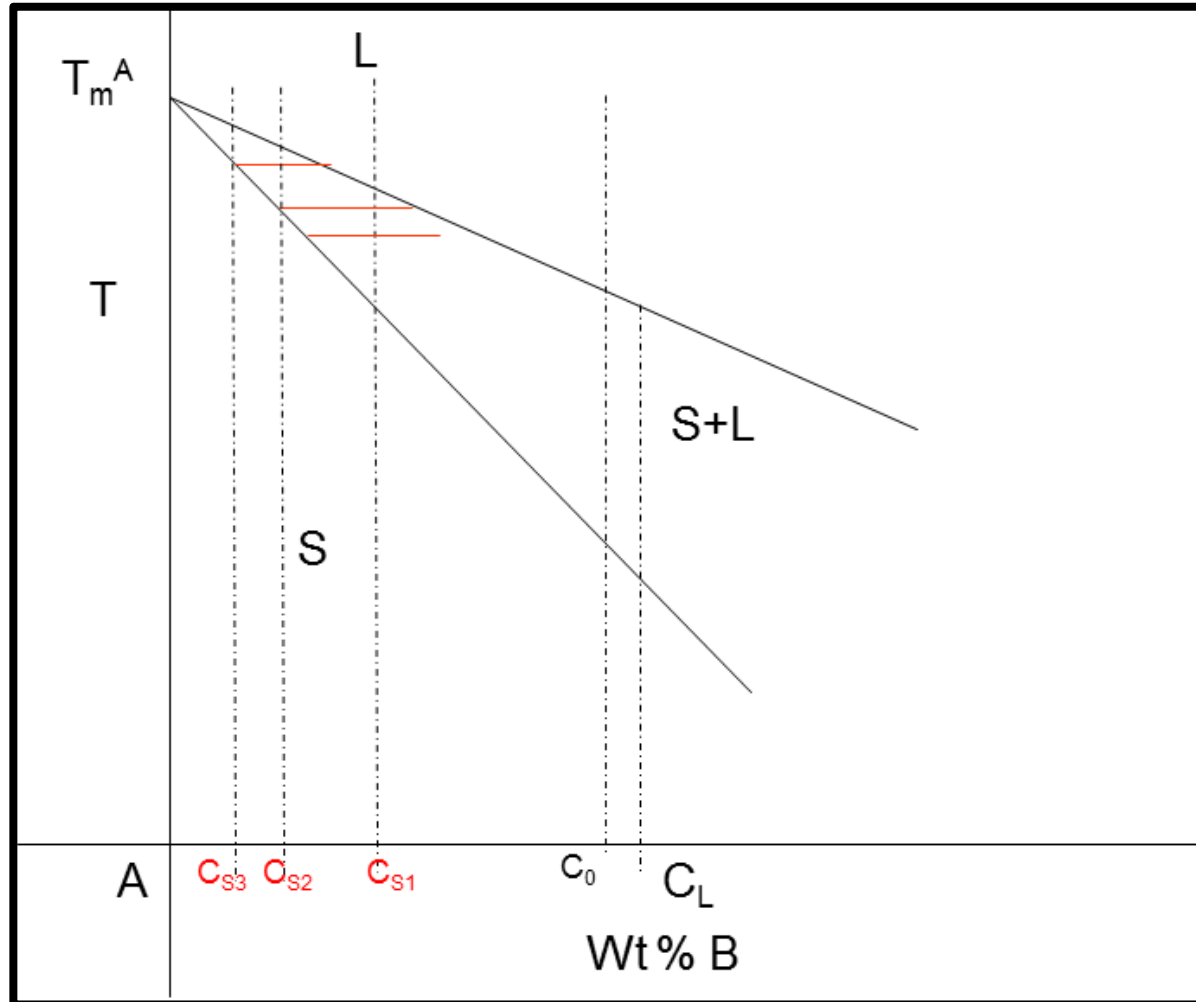
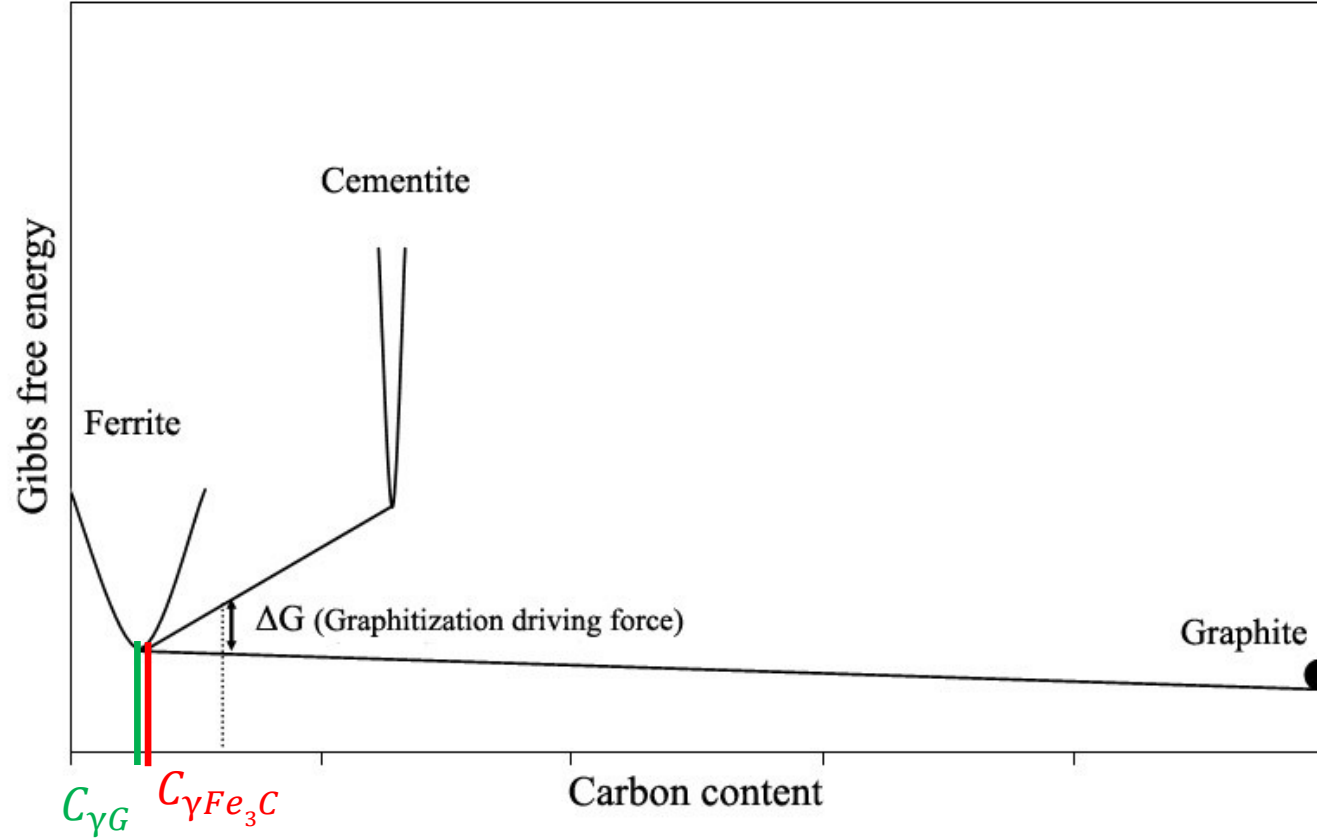


Figure: (a) First pass results in solid of C_{s1} composition at extreme left. 2nd pass gives C_{s2} , third pass gives C_{s3} and so on.
 (b) Effect of number of passes (n) on solute concentration in a solid bar

The Iron-Carbon system

G Vs. X diagram



Fe_3C is the metastable phase, having high temperature

Graphite (C) is the stable phase, having low temperature

Similarly,

$$C_{\alpha G} < C_{\alpha Fe_3C}$$

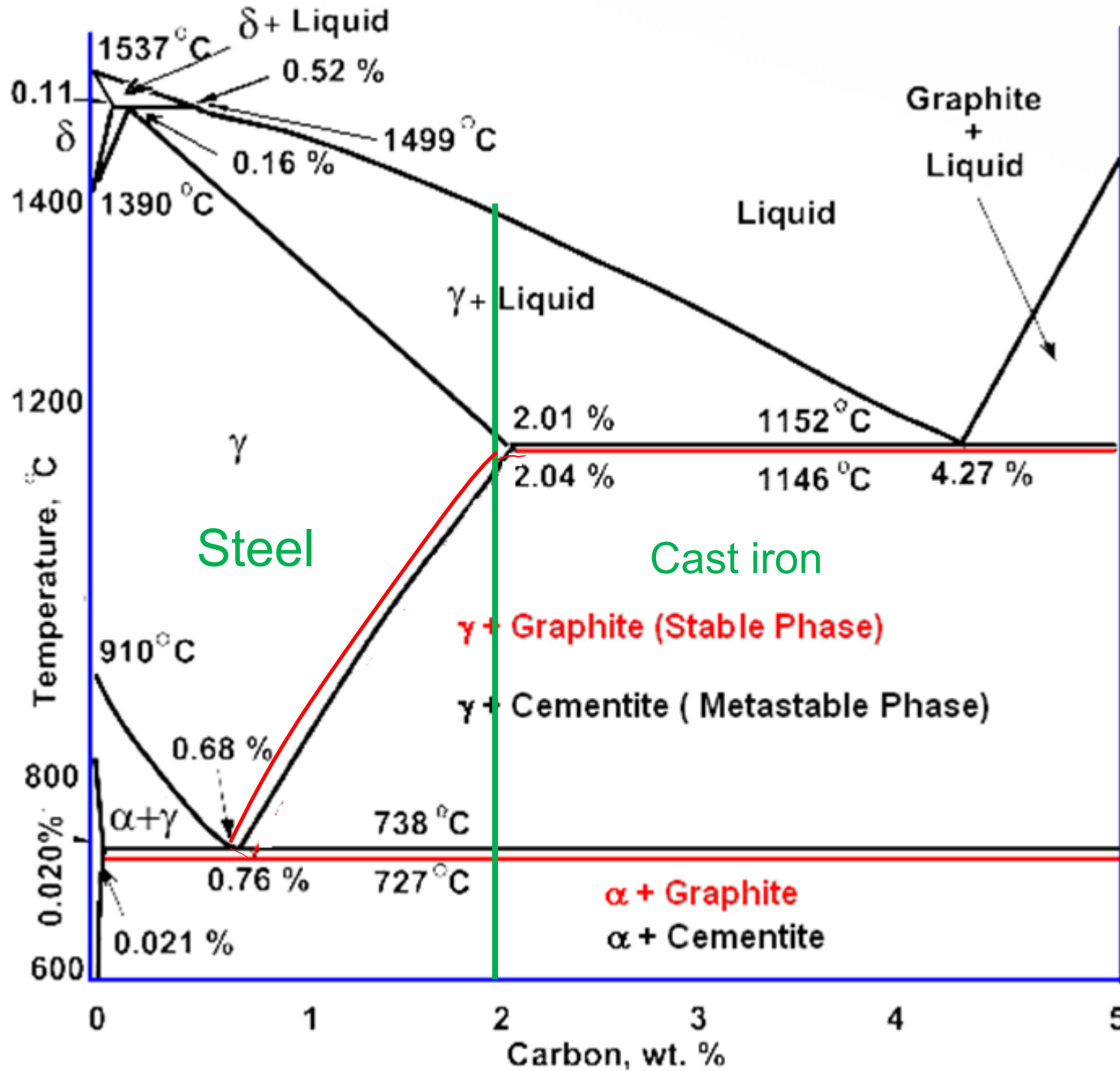
$$C_{LG} < C_{L Fe_3C}$$

$C_{\gamma G}$: Composition of γ -Fe in equilibrium with graphite (G)

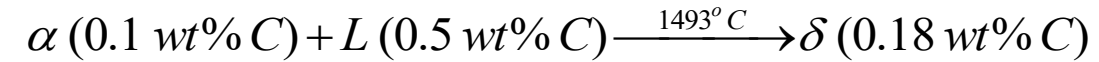
$C_{\gamma Fe_3C}$: Composition of γ -Fe in equilibrium with cementite (Fe_3C)

$$C_{\gamma G} < C_{\gamma Fe_3C}$$

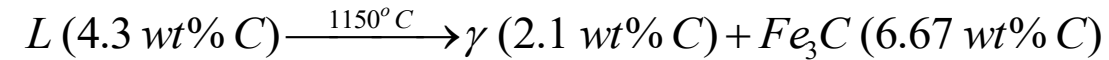
The Iron-Carbon system



Peritectic Reaction



Eutectic Reaction



Eutectoid Reaction



Thermodynamic driving force for a phase transformation

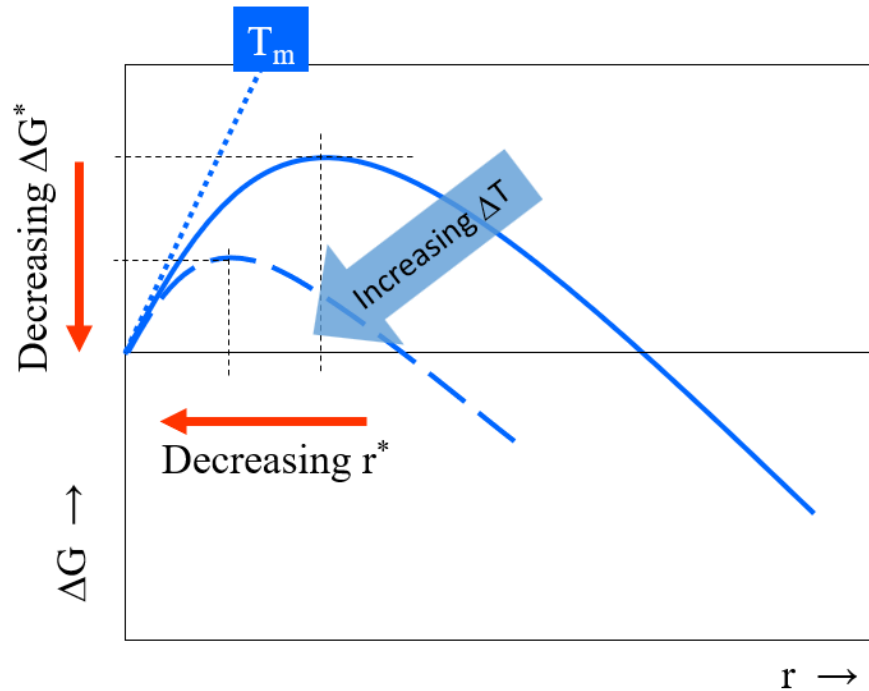
Decrease in Gibbs free energy

Liquid → solid

$$\Delta G_V = -ve$$

- *Atoms execute a Brownian motion, dance around, make and break friends with other atoms and finally rest in peace in the extremum energy configuration of **stable critical size nuclear family** (or stable family)*
- A change of any material must go through a nucleation and growth process: **Fundamental Law of Nature**

Nucleation



Total change in free energy:

$$\Delta G = -\frac{4}{3}\pi r^3 \Delta G_V + 4\pi r^2 \gamma_{SL}$$

Liquid(L)-Solid(S) transformation:

$$r^* = \frac{2 \gamma_{SL}}{\Delta G_V} = \frac{2 \gamma_{SL} T_m}{\Delta H \cdot \Delta T}$$

$$\Delta G^* = \frac{16}{3} \pi \frac{\gamma_{SL}^3}{\Delta G_V^2} = \frac{16}{3} \pi \frac{\gamma_{SL}^3 T_m^2}{\Delta H^2 \Delta T^2}$$

Thank you